

Operators manual for Attachment



Front loader for Avant 200 and 250 series tractors
Working width 1000 mm
1200 mm
1400 mm
Working depth 200 mm, 250 mm, 300 mm

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1. Introduction

Avant Tecno Oy wants to thank you for purchasing this original Avant attachment. It is the result of Avant's long experience in design and manufacturing of attachments. We ask you that you read and understand the contents of this manual completely before operating the attachment. This will improve your operating and maintenance efficiency, help avoid breakdowns and damage and extend your machines life. Do not hesitate to contact your local Avant dealer for advice regarding service, parts and any problems relating to the operation of your attachment.

2. Designed purpose

Flail mowers (A21030 and A2853) are intended for Avant 200 and 500 series loaders only. It is designed for cutting down tall grass, hay, shrubs or similar vegetation. The flails of the mower are Y-shaped for optimum mulching performance. These "Y" shaped blades give more mowing power, but because of them the mowed surface is not completely even. This is why the flail mower should not be used for mowing yard lawns.

For mowing yard lawns Avant has developed several horizontal lawn mowers.

3. Safety instructions



- Never use the flail mower without instructions. Read all the decals and this manual.
- Start the operation slowly and carefully.
- Do not wear loose clothing, long uncovered hair or jewellery near the machine.
- When coupling the attachment, make sure that the attachment locking pins lock in positively.
- Keep hands, feet and clothing away from any moving part.
- Make sure, that there are no bystanders in the working area before starting the flail mower. **The safety area is 35 meters or 100 feet.**
- Do not leave the cockpit while the mower is running.
- Never go under an uplifted attachment.
- Always carry the attachment as low as possible, because of the optimal weight distribution.
- Never perform **any** service operations when the engine is running.

4. Technical specification

A21030:

Operating width:	1250 mm
Total width:	1420 mm
Height:	670 mm
Length:	1400 mm
Weight:	250 kg
Amount of flails:	60 pcs

A2853:

Operating width:	1000 mm
Total width:	1170 mm
Height:	670 mm
Length:	1400 mm
Weight:	210 kg
Amount of flails:	48 pcs

- The mower is equipped with a floating attachment plate (the attachment plate moves freely up and down following the contours of the ground).
- The cutting height of the mower can be adjusted with plastic spacers around the axels of the support wheels.

5. Attaching the flail mower to the loader

5.1 Mechanical attachment

Coupling the attachment to a 500 series loader happens as follows:

Stage 1:

- Lift up the two locking pins on either side of the attachment coupling plate and turn them backward so that they remain in the locked up position.
- Make sure, that the pins remain in the up position; otherwise you cannot couple the attachment properly!



Stage 2:

- Turn the attachment coupling plate with the tilt movement so that the upper edge of the plate leans forward.
- Drive the loader into the attachment.



Stage 3:

- Lift the boom a little so that the attachment lifts off of the ground.
- Pull the boom control lever to the left so that the lower edge of the attachment coupling plate turns into the attachment.

**Stage 4:**

- Pull the boom control lever more to the left so that attachment coupling plate turns more and the bolts on the boom push the pins down in the holes of the attachment.
- **CAUTION! Make sure that the pins lock properly down in the holes of the attachment.**



Always remember the safety instructions from paragraph 3, when attaching the flail mower.

Coupling the attachment to a 200 series loader happens as follows:

Stage 1:

- Lift up the locking pin and turn it backward so that it remains in the locked up position.
- Make sure that the pin remains in the up position, otherwise you cannot couple the attachment properly!



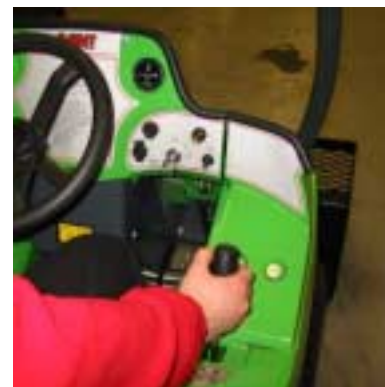
Stage 2:

- Turn the attachment coupling plate with the tilt movement so that the upper edge of the plate leans forward.
- Drive the loader into the attachment so that the pins in the upper left and right corner of the attachment coupling plate go into the holes of the attachment bracket.



Stage 3:

- Lift the boom a little so that the attachment lifts off the ground.
- Pull the boom control lever to the left so that the lower edge of the attachment coupling plate turns into the attachment



Stage 4:

- Pull the boom control lever more to the left so that the attachment coupling plate turns more and the boom pushes the pin down in the hole of the attachment.
- **CAUTION! Make sure that the pin locks properly down in the hole of the attachment.**



5.2 Connecting the hydraulic couplings

Coupling the 500 series loader

The flail mower has three hydraulic quick couplings, which should be attached to the corresponding couplings in the loader. There are two male couplings and one female coupling in the mower. The loader has correspondingly one male coupling and two female couplings.

The hoses from the mower should be attached to the loader correctly (pressure line to pressure line and return line to return line). The blue rubber caps of the couplings always mark the pressure line (one on the loader and one on the attachment). Always connect them as a pair. This way the flails start to rotate in the right direction, when the loader's auxiliary hydraulics control lever is in its locked position.



If the flail mower starts to rotate in the wrong direction when the loader's auxiliary hydraulics control lever is turned to right in its locked position, please change the coupling of the return- and pressure hose. **See picture 1**



Picture 1



Always **make sure**, that **all couplings are properly locked in before starting the engine**. The hydraulic motor of the mower may get damaged, if one of the hoses is not locked in.



Make sure that the hydraulic hoses **do not get overstretched** in any boom or attachment plate position and are **not in a position** where they can be **trapped during the operation** of the machine and attachment.

- For coupling the hydraulic hoses see also the operators manual of the loader.
- Always remember the safety instructions from paragraph 3, when coupling the attachments.

Coupling the 200 series loader

The mower has three hydraulic quick couplings, which should be attached to the corresponding couplings in the loader. There are two male couplings and one female coupling in the mower. The loader has correspondingly one male coupling and two female couplings.

The hoses from the flail mower should be attached to the loader correctly (pressure line to pressure line and return line to return line). The red caps of the couplings always mark the pressure line (one on the loader and one on the attachment). This way the flails start to rotate in the right direction, when the loaders auxiliary hydraulics pedal is pressed down to its locked position.

The 200 series loaders and their attachments are equipped with flat face couplings, because they are more convenient and clean in everyday use. However there are some things you have to remember when coupling and uncoupling a flat face coupling.

1. Always try to **keep the couplings as clean as possible**. Dirt, ice etc might prevent the coupling from functioning properly.
2. Although the joint surface seems flat there are sprung parts in the coupling, which give in when pressed together correctly. If the couplings do not lock in to each other try **to release pressure by moving the auxiliary hydraulics pedal up and down** a couple of times and then try again.
3. **Before uncoupling the couplings always stop the engine** from of the loader and **move the auxiliary hydraulics pedal up and down** a couple of times. This releases any pressure left to the attachment and ensures easy coupling the next time you are using the attachment.
4. Flat face couplings have a locking mechanism, which prevents them to uncouple accidentally. When removing them from each other make sure that the ball and the slot from the female coupling are facing each other. **See picture 2**



Picture 2



Always **make sure**, that **all couplings are properly locked in before starting the engine**. The hydraulic motor of the mower may get damaged, if one of the couplings is not locked in.



If the female coupling will not release from the male coupling when its collar is pulled backwards, rotate the collar until the slot in it is facing the ball (**see picture 2**) and try again.



If the flails start to rotate in the wrong direction when the loader's auxiliary hydraulics pedal is pressed down to its locked position, please change the coupling of the return- and pressure hose. **See picture 3**



Picture 3



Make sure that the hydraulic hoses **do not get overstretched** in any boom or attachment plate position and are **not in a position** where they can be **trapped during the operation** of the machine and attachment.

- For coupling the hydraulic hoses see also the operators manual of the loader.
- Always remember the safety instructions from paragraph 3, when coupling the attachments.

6. Instructions for use

6.1 Setting the floating attachment plate

The flail mower is equipped with a floating attachment plate, which allows it to freely follow the contours of the ground. If the attachment plate on the mower would be rigid the mower would always lift the front wheels of the loader in the air on sudden bumps and the mower would be lifted in the air on sudden down hills.

To get the full advantage from this system you need to adjust it correctly before you start working with the mower. The floating should be set so, that it is in the middle of the floating slot. **See picture 4** This way the flail mower can move freely up and down as much as possible.



Picture 4

If your loader is equipped with boom floating you should switch it on as well. Loaders boom floating releases the boom so that it can move freely up and down. This allows the flail mower to move even more in case the brushed ground is uneven. The boom floating is switched on from its control lever in the loaders foot well. It should be switched on only when the boom is in its lowest position.

5.2 Adjusting the cutting height

The cutting height of the flail mower can be adjusted with plastic rings around the axel of the wheel supports.

Adjusting is done as follows:

1. Lift the mower up with your loader and turn off the engine. **NOTE: Do not go under the uplifted mower!**
2. Open and remove the pin from the vertical axel of the wheel support. **(Picture 5)**
3. Pull the wheel support away from its bracket.
4. Move the plastic rings to the upper side of the bracket if you wish to lower the mower and other way round if you wish to increase cutting height. **(See picture 6)**
5. Push the vertical wheel support axel back trough the bracket and put the pin back on.
6. Repeat phases 2 – 5 with all the other wheels as well. When you are finished, make sure, that there are equal amount of same sized plastic rings on top of each bracket.



Picture 5



Picture 6

7. General instructions

After the flail mower is attached to the loader's quick attachment plate and the hydraulic couplings are attached you are ready for work. Before turning the mower on, make sure that there are no bystanders in the working area. Turning the auxiliary hydraulics lever right to its lock position turns the mower on (In 200 series press the auxiliary hydraulics pedal down to its lock position)

When you are working with the flail mower you should keep the engine revolutions at full in order to get the best possible mowing result. If you have a plus-model loader (520+, 523+, 528+ or 530+) you should operate the flail mower with both pumps. (2-Pump selection on the loaders dashboard, see loaders operators manual) If the mower does not cut perfectly even when both pumps are activated try to slow the driving speed, still keeping the engine revolutions to the maximum.

8. Service

8.1 Greasing points

For greasing you need a grease press and a tube of grease, which will fit to the press. Both of these are available in a regular hardware store.

The flail mower has four (4) greasing points. One on the bearing of the cutting axle (on the left side of the mower, when viewed from the front) and three on the vertical axle of the support wheels.

The cutting axle bearing should be greased after every 7-10 working hours and the support wheels axles should be greased approximately twice a year. Do not add more than a couple of presses from the grease press per one overhaul.

By following these greasing instructions you will increase the life of your flail mower significantly.

8.2 Changing and sharpening the flails

The flail mower has 48 or 60 blades depending on the model. It is normal that the blades get dull or wear out during the use. If the blade is evenly dull it can be sharpened, but if there are large dents or such the blades should be changed.

Both of the flail mowers use the same blade. Product number for the blade is **A44354**

9. Warranty terms for Avant flail mower

(Product number A21030 for 1250 mm wide)

(Product number A2853 for 1000 mm wide)

During the first year of operation Avant Tecno Oy warrants to replace any part or repair any defect, which may occur, subject to the terms below:

1. The product has received regular maintenance in accordance with schedules given by the manufacturer.
2. Any damage caused by operation in a negligent manner or exceeding the approved specifications detailed in this manual is excluded.
3. Avant Tecno Oy accepts no responsibility for interruption to working or any other consequential losses resulting from any failure of the product.
4. Only Avant Tecno Oy approved replacement or original quality parts shall be used during routine maintenance.
5. The Avant Warranty excludes any consumable parts (e.g. chains or bearings) except where it can be clearly shown that these parts were defective on original supply.

In the event a fault occurs which is attributable to manufacturing or assembly defect you should arrange to return your attachment to your authorized dealer for repair. Travel and freight costs are excluded.